Yitao Zeng 曾亦陶

zengyt2023@mail.sustech.edu.cn | GitHub | Website | Google Scholar

Education

Southern University of Science and Technology (SUSTech) 2023 – Expected Jun 2027
B.Eng. in Automation **TOEFL: 110/120** **GPA: 3.87/4.0 (Rank: 2/35)**

University of Toronto Jan 2027 – Apr 2027 (upcoming)
Incoming Engineering Exchange Student (Selected as 1 of 2 university-wide nominees)

Projects & Research

Project demos and presentation slides are available on my website.

Aerial Grasping with a Physical UAV Jun 2026 – Present
SUSTech — Advisor: Boyu Zhou, Assistant Professor; manuscript in preparation

- Led the deployment of an RL-based aerial-grasping UAV. The system grasped a stationary plastic bottle in 7 of 10 trials at a grasping flight speed of 1.8 m/s, using onboard perception from a D455 depth camera.
- Designed a target-state estimation pipeline that temporally aligns 6D object-pose estimates with VINS states, integrating pose propagation and Kalman filtering. Investigated localization bias through comparisons between rosbag recordings and motion-capture ground truth.
- Diagnosed gripper-induced flight-policy degradation that stalled training; implemented a three-stage RL curriculum: flight adaptation to gripper motion, teacher-supervised closure, and policy-controlled grasping with progressively tighter tolerances.
- Proposed and integrated time-shifted teacher supervision to account for gripper-closing delay during deployment and enable anticipatory closure.

Imitation Learning for Robotic Manipulation Feb 2025 – Present

1. 3D Diffusion Policy Feb 2025 – Dec 2025

SUSTech — Course Instructor: Max Q.-H. Meng, Chair Professor

- Owned point-cloud preprocessing with workspace cropping and farthest-point sampling for a team-built 3D Diffusion Policy pipeline; participated in camera calibration, demonstration collection, policy training, and ROS deployment.
- Evaluated four tasks (20 trials each): bowl picking (70%), drawer closing (100%), lemon placement in a bowl (30%), and a lemon–drawer sequence (10%). Multi-step execution remained unreliable.

2. Iron Bestie: Agent-Driven Robotic Manipulation Aug 2026 – Present

- A five-person hackathon project that combined agent orchestration, speech interaction, visual perception, and robotic-arm skills for cosmetics organization and personalized assistance.
- Led NERO arm assembly and networking, demonstration collection, ACT training and deployment, and MCP–Agent integration. Coordinated team tasks and delivered the main presentation in English.

Reinforcement Learning for Bipedal and Wheel-Legged Locomotion Jul 2025 – Jan 2026
LimX Dynamics TRON Camp

- Trained a point-foot locomotion policy in Isaac Sim and evaluated sim-to-sim transfer in MuJoCo for the preliminary stage. Tested traversal over uneven terrain, slopes, and stairs under external disturbances.
- In follow-up point-foot simulation experiments, implemented direction-decoupling reward terms and evaluated velocity-tracking errors and responses to external pushes.
- Independently adapted and deployed a locomotion policy on a wheel-legged robot for the finals, completing five successful climbs in five trials on an 18 cm physical stair as a bonus challenge.

Industrial Research Experience

Robot Learning, Foundation-Model Deployment, and Data Curation Jan 2026 – Jun 2026
Galbot — Embodied Intelligence Algorithm Department
Research group led by He Wang, Assistant Professor, Peking University

- Surveyed depth-aware VLA policies and selected DP3 in response to StereoVLA reviewer feedback. Integrated a Dockerized ROS–Franka inference stack with ZED sensing and ZeroMQ communication for StereoVLA baseline adaptation and deployment support.
- Developed a parallel TFDS-to-LeRobot conversion pipeline for 5 million input SynGrasp-1B trajectories, achieving a 99.5% trajectory conversion success rate and reducing storage for the same dataset from 20 TB to 5 TB (75%).
- Restored the robot motion-control system for real-world foundation-model deployment; analyzed initial poses in the simulation dataset to guide real-robot grasping experiments, achieving 10/10 successful grasps from selected starting configurations.
- Reviewed robot-learning data curation methods and attempted to reproduce CUPID-based data selection for clothes folding, examining limitations related to task complexity and dataset suitability.

Publications

StereoVLA: Enhancing Vision-Language-Action Models with Stereo Vision
Shengliang Deng*, Mi Yan*, Yixin Zheng*, Jiayi Su, Wenhao Zhang, **Yitao Zeng**, . . . , He Wang
Accepted at RSS 2026; co-author of the updated arXiv manuscript

MirrorSentinel: Suppressing Mirror-Induced Ghost Structures in Elevator LiDAR Maps with Visual Depth
Yugong Wang, Shengzhe Gan, Yu Zhan, Zhuoli Ouyang, Chuyao Fu, **Yitao Zeng**, Hong Zhang
Accepted to IEEE International Conference on Robotics and Biomimetics (ROBIO), 2026

PerchRL: Vision-Based Agile Perching on Inclined Platforms under Rapid and Irregular Motion
Zihong Lu, Zongzhuo Liu, Huaxu Li, **Yitao Zeng**, . . . , Boyu Zhou
Submitted to IEEE International Conference on Robotics and Automation (ICRA), 2027

Selected Awards & Honors

- **Grand MVP Award, Hong Kong Physical AI Hackathon** (2026) — Highest award among 12 finalist teams; personally interviewed and featured by the South China Morning Post.
- **Guangdong Provincial Gold Award, China International College Students' Innovation Competition** (2025) — Team award; one of 136 provincial Gold projects in a Ministry of Education-led national competition. [SUSTech Award Report] [Project Coverage].
- **Outstanding Team Award, LimX Dynamics TRON Camp** (2025) — Sole Outstanding Team Award among seven finalist teams for reinforcement learning-based robot locomotion.
- **University Motto Scholarship – Rixin (Innovation & Entrepreneurship) Nomination Award, SUSTech** (2026) — Shuren College nominee; one of 8 university-wide finalists in this category. [Criteria]
- **First Prize Scholarship, SUSTech** (2024, 2026) — top 5%.
- **Second Prize Scholarship, SUSTech** (2025) — top 10%.
- **Platinum Award & Best Functional Design Award, SUSTech Da Vinci Challenge Camp** (2024) — 1st among six competing teams in an interdisciplinary engineering design and prototyping challenge.